

CALCULATION OF COEFFICIENTS APPEARING IN EXPANSIONS OF ZONAL POLYNOMIALS OF HERMITIAN MATRICES

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Summary: Zonal polynomials of a hermitian matrix are
expressed in terms of its characteristic roots
and functions of its roots. These expressions
are used to calculate the coefficients appearing
in certain expansions of zonal polynomials.

1. INTRODUCTION

As zonal polynomials of real symmetric matrices play an important role in the real multivariate statistical theory, zonal polynomials of hermitian matrices also play an important role in the multivariate complex statistical theory. James (1964) gives a brief description of zonal polynomials of hermitian matrices.

Certain expansions of zonal polynomials and products of zonal polynomials of hermitian matrices are needed to derive certain

probability density functions of hermitian matrix variates and their roots. In this paper a certain property of zonal polynomials of hermitian matrices, given by James (1964), is used to express the zonal polynomials of a hermitian matrix in terms of:

- (i) its characteristic roots,
- (ii) monomial symmetric functions of its characteristic roots, and
- (iii) sums of g -th powers of its characteristic roots.

These expressions are tabulated and used to calculate the coefficients which appear in the expansions mentioned above. These coefficients are not the same as the coefficients which appear in the analogue expansions of zonal polynomials of real symmetric matrices.

2. DEFINITIONS AND PROPERTIES

Consider the following definitions:

Definition 2.1 (Macduffee (1946), p. 62)

The complex matrix $A: p \times p$, the kl -th element being $a_{kl} = a_{klR} + i a_{klI}$, is defined as a hermitian matrix if $A = \bar{A}'$.

From definition 2.1 follows that

$$a_{kl} = \bar{a}_{lk} = a_{lkR} - i a_{lkI},$$

$$a_{klR} = a_{lkR},$$

$$a_{klI} = -a_{lkI}, \quad (k, l=1, \dots, p).$$

Definition 2.2

The zonal polynomial $\tilde{C}_\kappa(A)$ is defined as the component of $(\text{tr}(A))^k$ in the irreducible invariant subspace V_κ which is the vector space of homogeneous polynomials $\phi(A)$ of degree k in the $n = \frac{1}{2}p(p+1)$ different complex elements of the hermitian matrix $A: p \times p$.

In definition 2.2 $\kappa = (k_1, \dots, k_p)$ is a partition of the integer k i.e. a set of integers $k_1 \geq \dots \geq k_p \geq 0$ such that $\sum_{i=1}^p k_i = k$; $V_k = \oplus V_\kappa$ and $(\text{tr}(A))^k$ has a unique decomposition in terms of zonal polynomials:

$$(2.1) \quad (\text{tr}(A))^k = \sum_{\kappa} \tilde{C}_\kappa(A).$$

The summation in (2.1) is taken over all partitions κ of k into not more than p parts. For $p=1$ follows that

$$(2.2) \quad \tilde{C}_\kappa(A) = \tilde{C}_\kappa(a) = a^k.$$

Definition 2.3 (James (1964), p. 481)

The i -th monomial symmetric function of the characteristic roots $\tilde{a}_1, \dots, \tilde{a}_p$ of $A: p \times p$ which are all real, is defined as

$$(2.3) \quad \alpha_i = \sum_{r_1 + \dots + r_p = i} \tilde{a}_1^{r_1} \dots \tilde{a}_p^{r_p} \quad \text{if } i=1, 2, \dots$$

$$= 1 \quad \text{if } i=0$$

$$= 0 \quad \text{if } i < 0$$

i.e. the sum of all monomials of $\tilde{a}_1, \dots, \tilde{a}_p$ of degree i .

James (1964) gives the following property of zonal polynomials of hermitian matrices which will be used to express $\tilde{C}(A)$ in terms of α_i , $i=1, \dots, k$.

Property (James (1964), p. 487-488)

For $A: p \times p$ hermitian:

$$(2.4) \quad \tilde{C}_\kappa(A) = x_{[\kappa]}(1) x_{\{\kappa\}}(A)$$

where $x_{[\kappa]}(1)$ is the dimension of the representation $[\kappa]$ of the symmetric group and is given as

$$(2.5) \quad x_{[\kappa]}(1) = \frac{k! \prod_{i < j}^p (k_i - k_j - i + j)}{\prod_{i=1}^p (k_i + p - i)!}$$

and $x_{\{\kappa\}}(A)$ is the character of the representation $\{\kappa\}$ of the linear group and is given as a symmetric function of the characteristic roots $\tilde{a}_1, \dots, \tilde{a}_p$ of $A: p \times p$:

$$(2.6) \quad x_{\{\kappa\}}(A) = \frac{|\tilde{a}_i^{k_j + p - j}|}{|(\tilde{a}_i^{p-j})|} = \frac{|\tilde{a}_j^{k_i + p - i}|}{|(\tilde{a}_j^{p-i})|} \\ = |(\alpha_{k_j - j + i})|$$

The expression given in (2.7) for $x_{\{\kappa\}}(A)$ is used to express $\tilde{C}_\kappa(A)$ in terms of α_i , ($i=1, \dots, k$). These formulae are given in table 2.1 for $k=1, 2, 3$ and 4. Table 2 of Farrel ((1980), p. 301-320) gives $x_{\{\kappa\}}(A)$ in terms of α_i , ($i=1, \dots, k$) for $k=1, 2, 3, 4$ and 5.

TABLE 2.1

Zonal polynomials of a hermitian matrix
in terms of α_i , ($i=1, \dots, k$)

k	κ	$\tilde{C}_\kappa(A)$
1	1	α_1
2	1^2	$\alpha_1^2 - \alpha_2$
	2	α_2
3	1^3	$\alpha_1^3 - 2\alpha_1\alpha_2 + \alpha_3$
	21	$2(\alpha_1\alpha_2 - \alpha_3)$
	3	α_3
4	1^4	$\alpha_1^4 - 3\alpha_1^2\alpha_2 + \alpha_2^2 + 2\alpha_1\alpha_3 - \alpha_4$
	21^2	$3(\alpha_1^2\alpha_2 - \alpha_2^2 - \alpha_1\alpha_3 + \alpha_4)$
	2^2	$2(\alpha_2^2 - \alpha_1\alpha_3)$
	31	$3(\alpha_1\alpha_3 - \alpha_4)$
	4	α_4

The contents of table 2.1 and definition 2.3 are used to express $\tilde{C}_\kappa(A)$ in terms of the characteristic roots of $A: p \times p$. These formulae for $\tilde{C}_\kappa(A)$ are given in table 2.2 for $p=1,2,3$ and 4 and $k=1,2,3$ and 4.

TABLE 2.2

Zonal polynomials of a hermitian matrix in terms
of the characteristic roots

p	k	κ	$C_{\kappa}(A)$	$C_{\kappa}(I_p)$
1	1	1	$\tilde{a}_1$	1
	2	2	$\tilde{a}_1^2$	1
	3	3	$\tilde{a}_1^3$	1
	4	4	$\tilde{a}_1^4$	1
2	1	1	$\tilde{a}_1 + \tilde{a}_2$	2
	2	1^2	$\tilde{a}_1 \tilde{a}_2$	1
		2	$\tilde{a}_1^2 + \tilde{a}_2^2 + \tilde{a}_1 \tilde{a}_2$	3
	3	21	$2(\tilde{a}_1^2 \tilde{a}_2 + \tilde{a}_1 \tilde{a}_2^2)$	4
		3	$\tilde{a}_1^3 + \tilde{a}_2^3 + \tilde{a}_1^2 \tilde{a}_2 + \tilde{a}_1 \tilde{a}_2^2$	4
	4	2^2	$2\tilde{a}_1^2 \tilde{a}_2^2$	2
		31	$3(\tilde{a}_1^3 \tilde{a}_2 + \tilde{a}_1 \tilde{a}_2^3 + \tilde{a}_1^2 \tilde{a}_2^2)$	9
3		4	$\tilde{a}_1^4 + \tilde{a}_2^4 + \tilde{a}_1^3 \tilde{a}_2 + \tilde{a}_1 \tilde{a}_2^3 + \tilde{a}_1^2 \tilde{a}_2^2$	5
	1	1	$\tilde{a}_1 + \tilde{a}_2 + \tilde{a}_3$	3
	2	1^2	$\tilde{a}_1 \tilde{a}_2 + \tilde{a}_1 \tilde{a}_3 + \tilde{a}_2 \tilde{a}_3$	3
		2	$\tilde{a}_1^2 + \tilde{a}_2^2 + \tilde{a}_3^2 + \tilde{a}_1 \tilde{a}_2 + \tilde{a}_1 \tilde{a}_3 + \tilde{a}_2 \tilde{a}_3$	6
	3	1^3	$\tilde{a}_1 \tilde{a}_2 \tilde{a}_3$	1
		21	$2(\tilde{a}_1^2 \tilde{a}_2 + \tilde{a}_1^2 \tilde{a}_3 + \tilde{a}_2^2 \tilde{a}_1 + \tilde{a}_2^2 \tilde{a}_3 + \tilde{a}_3^2 \tilde{a}_1 + \tilde{a}_3^2 \tilde{a}_2 + 2\tilde{a}_1 \tilde{a}_2 \tilde{a}_3)$	16

P.	k	κ	$\tilde{C}_\kappa(A)$	$\tilde{C}_\kappa(A)$
4	4	3	$\tilde{a}_1^3 + \tilde{a}_2^3 + \tilde{a}_3^3 + \tilde{a}_1^2\tilde{a}_2 + \tilde{a}_1^2\tilde{a}_3 + \tilde{a}_2^2\tilde{a}_1$ $+ \tilde{a}_2^2\tilde{a}_3 + \tilde{a}_3^2\tilde{a}_1 + \tilde{a}_3^2\tilde{a}_2 + \tilde{a}_1\tilde{a}_2\tilde{a}_3$	10
		21^2	$3(\tilde{a}_1^2\tilde{a}_2\tilde{a}_3 + \tilde{a}_1\tilde{a}_2^2\tilde{a}_3 + \tilde{a}_1\tilde{a}_2\tilde{a}_3^2)$	9
		2^2	$2(\tilde{a}_1^2\tilde{a}_2^2 + \tilde{a}_1^2\tilde{a}_3^2 + \tilde{a}_2^2\tilde{a}_3^2 + \tilde{a}_1^2\tilde{a}_2\tilde{a}_3$ $+ \tilde{a}_1\tilde{a}_2^2\tilde{a}_3 + \tilde{a}_1\tilde{a}_2\tilde{a}_3^2)$	12
		31	$3(\tilde{a}_1^3\tilde{a}_2 + \tilde{a}_1^3\tilde{a}_3 + \tilde{a}_2^3\tilde{a}_1 + \tilde{a}_2^3\tilde{a}_3 + \tilde{a}_3^3\tilde{a}_1$ $+ \tilde{a}_3^3\tilde{a}_2 + \tilde{a}_1^2\tilde{a}_2^2 + \tilde{a}_1^2\tilde{a}_3^2 + \tilde{a}_2^2\tilde{a}_3^2$ $+ 2\tilde{a}_1^2\tilde{a}_2\tilde{a}_3 + 2\tilde{a}_1\tilde{a}_2^2\tilde{a}_3 + 2\tilde{a}_1\tilde{a}_2\tilde{a}_3^2)$	45
		4	$\tilde{a}_1^4 + \tilde{a}_2^4 + \tilde{a}_3^4 + \tilde{a}_1^3\tilde{a}_2 + \tilde{a}_2^3\tilde{a}_1$ $+ \tilde{a}_2^3\tilde{a}_3 + \tilde{a}_3^3\tilde{a}_1 + \tilde{a}_3^3\tilde{a}_2 + \tilde{a}_1^2\tilde{a}_2^2$ $+ \tilde{a}_1^2\tilde{a}_3^2 + \tilde{a}_2^2\tilde{a}_3^2 + \tilde{a}_1^2\tilde{a}_2\tilde{a}_3 + \tilde{a}_1\tilde{a}_2^2\tilde{a}_3$ $+ \tilde{a}_1\tilde{a}_2\tilde{a}_3^2$	15
	1	1	$\tilde{a}_1 + \tilde{a}_2 + \tilde{a}_3 + \tilde{a}_4$	4
		1^2	$\tilde{a}_1\tilde{a}_2 + \tilde{a}_1\tilde{a}_3 + \tilde{a}_1\tilde{a}_4 + \tilde{a}_2\tilde{a}_3 + \tilde{a}_2\tilde{a}_4$ $+ \tilde{a}_3\tilde{a}_4$	6
		2	$\tilde{a}_1^2 + \tilde{a}_2^2 + \tilde{a}_3^2 + \tilde{a}_4^2 + \tilde{a}_1\tilde{a}_2 + \tilde{a}_1\tilde{a}_3$ $+ \tilde{a}_1\tilde{a}_4 + \tilde{a}_2\tilde{a}_3 + \tilde{a}_2\tilde{a}_4 + \tilde{a}_3\tilde{a}_4$	10
		1^3	$\tilde{a}_1\tilde{a}_2\tilde{a}_3 + \tilde{a}_1\tilde{a}_3\tilde{a}_4 + \tilde{a}_1\tilde{a}_2\tilde{a}_4 + \tilde{a}_2\tilde{a}_3\tilde{a}_4$	4
	3	21	$2(\tilde{a}_1^2\tilde{a}_2 + \tilde{a}_1^2\tilde{a}_3 + \tilde{a}_1^2\tilde{a}_4 + \tilde{a}_2^2\tilde{a}_1 + \tilde{a}_2^2\tilde{a}_3$ $+ \tilde{a}_2^2\tilde{a}_4 + \tilde{a}_3^2\tilde{a}_1 + \tilde{a}_3^2\tilde{a}_2 + \tilde{a}_3^2\tilde{a}_4$ $+ \tilde{a}_4^2\tilde{a}_1 + \tilde{a}_4^2\tilde{a}_2 + \tilde{a}_4^2\tilde{a}_3 + 2\tilde{a}_1\tilde{a}_2\tilde{a}_3$ $+ 2\tilde{a}_1\tilde{a}_2\tilde{a}_4 + 2\tilde{a}_2\tilde{a}_3\tilde{a}_4 + 2\tilde{a}_1\tilde{a}_3\tilde{a}_4)$	40

p	k	κ	$\tilde{C}_\kappa(A)$	$\tilde{C}_\kappa(I_p)$
		3	$\begin{aligned} &\tilde{a}_1^3 + \tilde{a}_2^3 + \tilde{a}_3^3 + \tilde{a}_4^3 + \tilde{a}_1^2\tilde{a}_2 + \tilde{a}_1^2\tilde{a}_3 \\ &+ \tilde{a}_1^2\tilde{a}_4 + \tilde{a}_2^2\tilde{a}_1 + \tilde{a}_2^2\tilde{a}_3 + \tilde{a}_2^2\tilde{a}_4 + \tilde{a}_3^2\tilde{a}_1 \\ &+ \tilde{a}_3^2\tilde{a}_2 + \tilde{a}_3^2\tilde{a}_4 + \tilde{a}_4^2\tilde{a}_1 + \tilde{a}_4^2\tilde{a}_2 + \tilde{a}_4^2\tilde{a}_3 \\ &+ \tilde{a}_1\tilde{a}_2\tilde{a}_3 + \tilde{a}_1\tilde{a}_2\tilde{a}_4 + \tilde{a}_2\tilde{a}_3\tilde{a}_4 + \tilde{a}_1\tilde{a}_3\tilde{a}_4 \end{aligned}$	20
	4	1^4	$\tilde{a}_1\tilde{a}_2\tilde{a}_3\tilde{a}_4$	1
		21	$\begin{aligned} &3(\tilde{a}_1^2\tilde{a}_2\tilde{a}_3 + \tilde{a}_1^2\tilde{a}_2\tilde{a}_4 + \tilde{a}_1^2\tilde{a}_3\tilde{a}_4 + \tilde{a}_1\tilde{a}_2^2\tilde{a}_3 \\ &+ \tilde{a}_2^2\tilde{a}_3\tilde{a}_4 + \tilde{a}_1\tilde{a}_2^2\tilde{a}_4 + \tilde{a}_1\tilde{a}_2\tilde{a}_3^2 \\ &+ \tilde{a}_1\tilde{a}_3^2\tilde{a}_4 + \tilde{a}_2\tilde{a}_3^2\tilde{a}_4 + \tilde{a}_1\tilde{a}_2\tilde{a}_4^2 \\ &+ \tilde{a}_1\tilde{a}_3\tilde{a}_4^2 + \tilde{a}_2\tilde{a}_3\tilde{a}_4^2 + 3\tilde{a}_1\tilde{a}_2\tilde{a}_3\tilde{a}_4) \end{aligned}$	45
		2^2	$\begin{aligned} &2(\tilde{a}_1^2\tilde{a}_2^2 + \tilde{a}_1^2\tilde{a}_3^2 + \tilde{a}_1^2\tilde{a}_4^2 + \tilde{a}_2^2\tilde{a}_3^2 + \tilde{a}_2^2\tilde{a}_4^2 \\ &+ \tilde{a}_3^2\tilde{a}_4^2 + \tilde{a}_1\tilde{a}_2\tilde{a}_3^2 + \tilde{a}_1\tilde{a}_2\tilde{a}_4^2 + \tilde{a}_1\tilde{a}_3\tilde{a}_4^2 \\ &+ \tilde{a}_1\tilde{a}_2^2\tilde{a}_3 + \tilde{a}_2^2\tilde{a}_3\tilde{a}_4 + \tilde{a}_1\tilde{a}_2^2\tilde{a}_4 \\ &+ \tilde{a}_1\tilde{a}_2\tilde{a}_3^2 + \tilde{a}_1\tilde{a}_3^2\tilde{a}_4 + \tilde{a}_2\tilde{a}_3^2\tilde{a}_4 \\ &+ \tilde{a}_1\tilde{a}_2\tilde{a}_4^2 + \tilde{a}_1\tilde{a}_3\tilde{a}_4^2 + \tilde{a}_2\tilde{a}_3\tilde{a}_4^2 \\ &+ 2\tilde{a}_1\tilde{a}_2\tilde{a}_3\tilde{a}_4) \end{aligned}$	40
		31	$\begin{aligned} &3(\tilde{a}_1^3\tilde{a}_2 + \tilde{a}_1^3\tilde{a}_3 + \tilde{a}_1^3\tilde{a}_4 + \tilde{a}_2^3\tilde{a}_1 + \tilde{a}_2^3\tilde{a}_3 \\ &+ \tilde{a}_2^3\tilde{a}_4 + \tilde{a}_3^3\tilde{a}_1 + \tilde{a}_3^3\tilde{a}_2 + \tilde{a}_3^3\tilde{a}_4 + \tilde{a}_4^3\tilde{a}_1 \\ &+ \tilde{a}_4^3\tilde{a}_2 + \tilde{a}_4^3\tilde{a}_3 + \tilde{a}_1^2\tilde{a}_2^2 + \tilde{a}_1^2\tilde{a}_3^2 + \tilde{a}_1^2\tilde{a}_4^2 \\ &+ \tilde{a}_2^2\tilde{a}_3^2 + \tilde{a}_2^2\tilde{a}_4^2 + \tilde{a}_3^2\tilde{a}_4^2 + 2(\tilde{a}_1^2\tilde{a}_2\tilde{a}_3 \\ &+ \tilde{a}_1^2\tilde{a}_2\tilde{a}_4 + \tilde{a}_1^2\tilde{a}_3\tilde{a}_4 + \tilde{a}_1\tilde{a}_2^2\tilde{a}_3 + \tilde{a}_2^2\tilde{a}_3\tilde{a}_4 \\ &+ \tilde{a}_1\tilde{a}_2^2\tilde{a}_4 + \tilde{a}_1\tilde{a}_2\tilde{a}_3^2 + \tilde{a}_1\tilde{a}_3^2\tilde{a}_4 + \tilde{a}_2\tilde{a}_3^2\tilde{a}_4 \\ &+ \tilde{a}_1\tilde{a}_2\tilde{a}_4^2 + \tilde{a}_1\tilde{a}_3\tilde{a}_4^2 + \tilde{a}_2\tilde{a}_3\tilde{a}_4^2) \\ &+ 3\tilde{a}_1\tilde{a}_2\tilde{a}_3\tilde{a}_4) \end{aligned}$	135

p	k	κ	$\tilde{C}_\kappa(A)$	$\tilde{C}_\kappa(I_p)$
		4	$ \begin{aligned} & \tilde{a}_1^4 + \tilde{a}_2^4 + \tilde{a}_3^4 + \tilde{a}_4^4 + \tilde{a}_1^3 \tilde{a}_2 + \tilde{a}_1^3 \tilde{a}_3 \\ & + \tilde{a}_1^3 \tilde{a}_4 + \tilde{a}_2^3 \tilde{a}_1 + \tilde{a}_2^3 \tilde{a}_3 + \tilde{a}_2^3 \tilde{a}_4 + \tilde{a}_3^3 \tilde{a}_1 \\ & + \tilde{a}_3^3 \tilde{a}_2 + \tilde{a}_3^3 \tilde{a}_4 + \tilde{a}_4^3 \tilde{a}_1 + \tilde{a}_4^3 \tilde{a}_2 + \tilde{a}_4^3 \tilde{a}_3 \\ & + \tilde{a}_1^2 \tilde{a}_2^2 + \tilde{a}_1^2 \tilde{a}_3^2 + \tilde{a}_1^2 \tilde{a}_4^2 + \tilde{a}_2^2 \tilde{a}_3^2 + \tilde{a}_2^2 \tilde{a}_4^2 \\ & + \tilde{a}_3^2 \tilde{a}_4^2 + \tilde{a}_1^2 \tilde{a}_2 \tilde{a}_3 + \tilde{a}_1^2 \tilde{a}_3 \tilde{a}_4 + \tilde{a}_1^2 \tilde{a}_2 \tilde{a}_4 \\ & + \tilde{a}_1 \tilde{a}_2^2 \tilde{a}_3 + \tilde{a}_1 \tilde{a}_2 \tilde{a}_4^2 + \tilde{a}_2^2 \tilde{a}_3 \tilde{a}_4 + \tilde{a}_1 \tilde{a}_2 \tilde{a}_3^2 \\ & + \tilde{a}_1 \tilde{a}_3^2 \tilde{a}_4 + \tilde{a}_2 \tilde{a}_3^2 \tilde{a}_4 + \tilde{a}_1 \tilde{a}_2 \tilde{a}_4^2 + \tilde{a}_1 \tilde{a}_3 \tilde{a}_4^2 \\ & + \tilde{a}_2 \tilde{a}_3 \tilde{a}_4^2 + \tilde{a}_1 \tilde{a}_2 \tilde{a}_3 \tilde{a}_4 \end{aligned} $	35

The contents of table 2.2 can be used to express $\tilde{C}_\kappa(A)$ in terms of the quantities

$$\begin{aligned}
a_g &= \text{sum of the } g\text{-th powers of the characteristic roots} \\
&\text{of } A: p \times p \\
&= \text{tr}(A^g).
\end{aligned}$$

These expressions of $\tilde{C}_\kappa(A)$ in terms of a_g are quite different from the contents of table 2 in Johnson and Kotz ((1972), p. 171) in which the zonal polynomials of a real symmetric matrix are expressed in terms of g -th powers of the characteristic roots of the real matrix. Table 3 of Farrel ((1980), p. 301-320) gives $x_{\{\kappa\}}(A)$ in terms of a_g for $k=1,2,3,4$ and 5. To compare the contents of table 3 of Farrel with the contents of table 2 of Johnson and Kotz the dimension of the representation $\{\kappa\}$ of the symmetric group, i.e. $x_{\{\kappa\}}(1)$, must be taken into account at Farrel's table.

3 EXPANSIONS INVOLVING ZONAL POLYNOMIALS OF HERMITIAN MATRICES

The product $\tilde{C}_\kappa(A) \tilde{C}_\tau(A)$ and the zonal polynomials $\tilde{C}_\kappa(I_p - A)$ and $\tilde{C}_\kappa(D_A^1)$ will be expressed underneath as expansions involving

zonal polynomials of $A: p \times p$ where

$$(3.1) \quad D_A^1 = \text{diag} [\tilde{a}_1, \dots, \tilde{a}_{p-1}, 1] .$$

In each case an example will be given to illustrate the calculation of the coefficients which appear in these expansions, by using the contents of tables 2.1 and 2.2. These coefficients are tabulated for small values of p and k .

3.1 The expansion of the product $\tilde{C}_\kappa(A) \tilde{C}_\tau(A)$

Let $A: p \times p$ be a hermitian matrix; then

$$(3.2) \quad \tilde{C}_\kappa(A) \tilde{C}_\tau(A) = \sum_{\delta} \tilde{g}_{\kappa, \tau}^{\delta} \tilde{C}_{\delta}(A)$$

where $\kappa \in P(k, p)$, $\tau \in P(r, p)$, $\delta \in P(k+t, p)$ and $P(k, p)$ is the set of partitions of the positive integer k into not more than p parts. The coefficients $\tilde{g}_{\kappa, \tau}^{\delta}$ are tabulated in table 3.1 for each partition of $d = k+t$ up to order 4.

Example 3.1

The coefficients $\tilde{g}_{\kappa, \tau}^{\delta}$ can easily be obtained by using the contents of table 2.1. The obtaining of the coefficients $\tilde{g}_{(1), (21)}^{(1^4)}$ in the expansion of $\tilde{C}_{(1)}(A) \tilde{C}_{(21)}(A)$ will be shown here.

From table 2.1 follows that

$$\begin{aligned} (3.3) \quad \tilde{C}_{(1)}(A) \tilde{C}_{(21)}(A) &= 2\alpha_1 (\alpha_2 \alpha_1 - \alpha_3) \\ &= 2\alpha_1^2 \alpha_2 - 2\alpha_1 \alpha_3 \\ &= 2(\alpha_1 \alpha_3 - \alpha_4) + 2(\alpha_2^2 - \alpha_1 \alpha_3) \\ &\quad + 2(\alpha_1^2 \alpha_2 - \alpha_2^2 - \alpha_1 \alpha_3 + \alpha_4) \\ &= \frac{2}{3} \tilde{C}_{(31)}(A) + \tilde{C}_{(2^2)}(A) + \frac{2}{3} \tilde{C}_{(21^2)}(A). \end{aligned}$$

Thus $\tilde{g}_{(1), (21)}^{(1^4)} = 0$, $\tilde{g}_{(1), (21)}^{(21^2)} = \frac{2}{3}$, $\tilde{g}_{(1), (21)}^{(2^2)} = 1$,
 $\tilde{g}_{(1), (21)}^{(31)} = \frac{2}{3}$ and $\tilde{g}_{(1), (21)}^{(4)} = 0$.

TABLE 3.1
The values of $\tilde{g}_{\kappa, \tau}^{\delta}$

$\tilde{c}_{\kappa}(\cdot) \tilde{c}_{\tau}(\cdot)$	d = 2		d = 3			d = 4				
	$\tilde{c}_{(2)}(\cdot)$	$\tilde{c}_{(1^2)}(\cdot)$	$\tilde{c}_{(3)}(\cdot)$	$\tilde{c}_{(21)}(\cdot)$	$\tilde{c}_{(1^3)}(\cdot)$	$\tilde{c}_{(4)}(\cdot)$	$\tilde{c}_{(31)}(\cdot)$	$\tilde{c}_{(2^2)}(\cdot)$	$\tilde{c}_{(21^2)}(\cdot)$	$\tilde{c}_{(1^4)}(\cdot)$
$\tilde{c}_{(1)}(\cdot) \tilde{c}_{(1)}(\cdot)$	1	1								
$\tilde{c}_{(1)}(\cdot) \tilde{c}_{(2)}(\cdot)$			1	$\frac{1}{2}$						
$\tilde{c}_{(1)}(\cdot) \tilde{c}_{(1^2)}(\cdot)$				$\frac{1}{2}$	1					
$\tilde{c}_{(1)}(\cdot) \tilde{c}_{(3)}(\cdot)$						1	$\frac{1}{3}$			
$\tilde{c}_{(1)}(\cdot) \tilde{c}_{(21)}(\cdot)$							$\frac{2}{3}$	1	$\frac{2}{3}$	
$\tilde{c}_{(1)}(\cdot) \tilde{c}_{(1^3)}(\cdot)$									$\frac{1}{3}$	1
$\tilde{c}_{(2)}(\cdot) \tilde{c}_{(2)}(\cdot)$						1	$\frac{1}{3}$	$\frac{1}{2}$		
$\tilde{c}_{(2)}(\cdot) \tilde{c}_{(1^2)}(\cdot)$							$\frac{1}{3}$		$\frac{1}{3}$	
$\tilde{c}_{(1^2)}(\cdot) \tilde{c}_{(1^2)}(\cdot)$								$\frac{1}{2}$	$\frac{1}{3}$	1

Remark 3.1

It is important to note that the coefficients $\tilde{g}_{k,r}^{\delta}$ are not identical to the coefficients $g_{k,r}^{\delta}$ tabulated in Khatri and Pillai ((1968), p. 218) and Gupta (1970) for $A:p \times p$ a real symmetric matrix.

3.2 Expansions of $\tilde{C}_k(I_p - A)$

Let $A:p \times p$ be a hermitian matrix; then

$$(3.4) \quad \tilde{C}_k(I_p - A) = \sum_{t=0}^k \sum_{\tau} \tilde{q}_{\tau} \tilde{C}_{\tau}(A).$$

The coefficients $\tilde{q}_{\tau}$ are tabulated in table 3.2 for $p=1,2,3$ and 4 and for each partition of $k=1,2$ and 3.

Example 3.2

The coefficients $\tilde{q}_{\tau}$ can easily be obtained by using the contents of table 2.2. The obtaining of $\tilde{q}_{\tau}$ in the expansion of $\tilde{C}_{(21)}(I_p - A)$ for $p=3$ will be shown here.

From table 2.2 follows for $p=3$ that

$$\begin{aligned} (3.5) \quad & \tilde{C}_{(21)}(I_p - A) \\ &= 2((1-\tilde{a}_1)^2(1-\tilde{a}_2) + (1-\tilde{a}_1)^2(1-\tilde{a}_3) + (1-\tilde{a}_2)^2(1-\tilde{a}_1) \\ & \quad + (1-\tilde{a}_2)^2(1-\tilde{a}_3) + (1-\tilde{a}_3)^2(1-\tilde{a}_1) + (1-\tilde{a}_3)^2(1-\tilde{a}_2) \\ & \quad + 2(1-\tilde{a}_1)(1-\tilde{a}_2)(1-\tilde{a}_3)) \\ &= 16 - 16(\tilde{a}_1 + \tilde{a}_2 + \tilde{a}_3) + 4(\tilde{a}_1^2 + \tilde{a}_2^2 + \tilde{a}_3^2 + 3\tilde{a}_1\tilde{a}_2 + 3\tilde{a}_1\tilde{a}_3 + 3\tilde{a}_2\tilde{a}_3) \\ & \quad - 2(\tilde{a}_1^2\tilde{a}_2 + \tilde{a}_1^2\tilde{a}_3 + \tilde{a}_2^2\tilde{a}_1 + \tilde{a}_3^2\tilde{a}_1 + \tilde{a}_3^2\tilde{a}_2 + 2\tilde{a}_1\tilde{a}_2\tilde{a}_3) \\ &= 16 - 16(\tilde{a}_1 + \tilde{a}_2 + \tilde{a}_3) + 8(\tilde{a}_1\tilde{a}_2 + \tilde{a}_1\tilde{a}_3 + \tilde{a}_2\tilde{a}_3) + 4(\tilde{a}_1^2 + \tilde{a}_2^2 + \tilde{a}_3^2 \\ & \quad + \tilde{a}_1\tilde{a}_2 + \tilde{a}_1\tilde{a}_3 + \tilde{a}_2\tilde{a}_3) - 2(\tilde{a}_1^2\tilde{a}_2 + \tilde{a}_1^2\tilde{a}_3 + \tilde{a}_2^2\tilde{a}_1 + \tilde{a}_3^2\tilde{a}_1 \\ & \quad + \tilde{a}_3^2\tilde{a}_2 + 2\tilde{a}_1\tilde{a}_2\tilde{a}_3) \\ &= \sum_{t=0}^3 \sum_{\tau} \tilde{q}_{\tau} \tilde{C}_{\tau}(A) \end{aligned}$$

where: $\tilde{q}_{(0)} = 16$, $\tilde{q}_{(1)} = -16$, $\tilde{q}_{(1^2)} = 8$, $\tilde{q}_{(2)} = 4$,
 $\tilde{q}_{(1^3)} = 0$, $\tilde{q}_{(21)} = -1$, $\tilde{q}_{(3)} = 0$.

TABLE 3.2
The values of $\tilde{q}_r$

k	p	κ	$\tilde{q}_{(0)}$	$\tilde{q}_{(1)}$	$\tilde{q}_{(1^2)}$	$\tilde{q}_{(2)}$	$\tilde{q}_{(1^3)}$	$\tilde{q}_{(21)}$	$\tilde{q}_{(3)}$
1	1	1	1	-1					
	2	1	2	-1					
	3	1	3	-1					
	4	1	4	-1					
2	1	2	1	-2		1			
	2	1^2	1	-1	1				
		2	3	-3		1			
	3	1^2	1	-2	1				
		21	6	-4		1			
	4	1^2	6	-3	1				
		2^2	10	-5		1			
3	1	1	1	-3		3			-1
	2	21	4	-6	6	2		-1	
		3	4	-6		4			-1
	3	1^3	1	-1	1		-1		
		21	16	-16	8	4		-1	
		3	10	-10		5			-1
	4	1^3	4	-4	2		-1		
		21	40	-30	10	6		-1	
		3	20	-15		6			-1

Let $A: p \times p$ be a hermitian matrix; then

$$(3.6) \quad \tilde{C}_k(I_p - A) = \tilde{C}_k(I_p) \sum_{t=0}^k \sum_{\tau} (-1)^t \tilde{q}_{k,\tau} \frac{\tilde{C}_{\tau}(A)}{\tilde{C}_{\tau}(I_p)}.$$

The coefficients $\tilde{q}_{k,\tau}$ are tabulated in table 3.3 for each partition of $k=1,2,3$ and 4.

Example 3.3

The coefficient $\tilde{q}_{k,\tau}$ can easily be obtained by using the contents of table 2.2. The obtaining of $\tilde{q}_{k,\tau}$ in the expansion of $\tilde{C}_{(21)}(I_p - A)$ for $p=3$ will be shown here.

From table 2.2 follows for $p=3$ that

$$\begin{aligned} (3.7) \quad & \frac{\tilde{C}_{(21)}(I_p - A)}{\tilde{C}_{(21)}(I_p)} \\ &= 1 - (\tilde{a}_1 + \tilde{a}_2 + \tilde{a}_3) + \frac{1}{4}(\tilde{a}_1^2 + \tilde{a}_2^2 + 3\tilde{a}_1\tilde{a}_2 + 3\tilde{a}_1\tilde{a}_3 + 3\tilde{a}_2\tilde{a}_3) \\ &\quad - \frac{1}{8}(\tilde{a}_1^2\tilde{a}_2 + \tilde{a}_1^2\tilde{a}_3 + \tilde{a}_2^2\tilde{a}_1 + \tilde{a}_2^2\tilde{a}_3 + \tilde{a}_3^2\tilde{a}_1 + \tilde{a}_3^2\tilde{a}_2 + 2\tilde{a}_1\tilde{a}_2\tilde{a}_3) \\ &= 1 - 3 \cdot \frac{1}{3}(\tilde{a}_1 + \tilde{a}_2 + \tilde{a}_3) + \frac{3}{2} \cdot \frac{1}{3}(\tilde{a}_1\tilde{a}_2 + \tilde{a}_1\tilde{a}_3 + \tilde{a}_2\tilde{a}_3) \\ &\quad + \frac{3}{2} \cdot \frac{1}{6}(\tilde{a}_1^2 + \tilde{a}_2^2 + \tilde{a}_3^2 + \tilde{a}_1\tilde{a}_2 + \tilde{a}_1\tilde{a}_3 + \tilde{a}_2\tilde{a}_3) \\ &\quad - 1 \cdot \frac{1}{8}(\tilde{a}_1^2\tilde{a}_2 + \tilde{a}_1^2\tilde{a}_3 + \tilde{a}_2^2\tilde{a}_1 + \tilde{a}_2^2\tilde{a}_3 + \tilde{a}_3^2\tilde{a}_1 + \tilde{a}_3^2\tilde{a}_2 + 2\tilde{a}_1\tilde{a}_2\tilde{a}_3) \\ &= \sum_{t=0}^3 \sum_{\tau} (-1)^t \tilde{q}_{k,\tau} \frac{\tilde{C}_{\tau}(A)}{\tilde{C}_{\tau}(I_p)} \end{aligned}$$

$$\text{where: } \tilde{q}_{(21),(0)} = 1, \quad \tilde{q}_{(21),(1)} = 3, \quad \tilde{q}_{(21),(1^2)} = \frac{3}{2},$$

$$\tilde{q}_{(21),(2)} = \frac{3}{2}, \quad \tilde{q}_{(21),(1^3)} = 0, \quad \tilde{q}_{(21),(21)} = 1,$$

$$\tilde{q}_{(21),(3)} = 0.$$

TABLE 3.3
The values of $\tilde{q}_{\kappa, \tau}$

k	κ	τ											
		0	1	1^2	2	1^3	21	3	1^4	21^2	2^2	31	4
1	1	1	1										
2	1^2	1	2	1									
	2	1	2		1								
3	1^3	1	3	3		1							
	21	1	3	$\frac{3}{2}$	$\frac{3}{2}$		1						
	3	1	3		3			1					
4	1^4	1	4	6		4			1				
	21^2	1	4	4	2	$\frac{4}{3}$	$\frac{8}{3}$			1			
	2^2	1	4	3	3		4				1		
	31	1	4	2	4		$\frac{8}{3}$	$\frac{4}{3}$				1	
	4	1	4		6			4					1

Remark 3.2

- (i) If $\tilde{C}_\kappa(I_p - A)$ is written in terms of the characteristic roots of $A: p \times p$ for different values of p by using the contents of table 2.2 it is easy to verify that the coefficients $\tilde{q}_{\kappa, \tau}$ remain the same, independently of the value of p , for given partitions κ of k and τ of t , $t \leq k$.
- (ii) It is important to note that the coefficients $\tilde{q}_{\kappa, \tau}$ are not identical to the coefficients $a_{\kappa, \tau}$ tabulated in Constantine ((1966), p. 224-225) for $A: p \times p$ a real symmetric matrix.

3.3 The expansion of $\tilde{C}_\kappa(D_A^1)$

Let $A: p \times p$ be a hermitian matrix; then

$$(3.8) \quad \tilde{C}_\kappa(D_A^1) = \sum_{t=0}^k \sum_{\tau} \tilde{b}_{\kappa, \tau} \tilde{C}_\kappa(D_A^*)$$

where D_A^1 is given by (3.1) and

$$(3.9) \quad D_A^* = \text{diag}[\tilde{a}_1, \dots, \tilde{a}_{p-1}].$$

The coefficients $\tilde{b}_{\kappa, \tau}$ are tabulated in table 3.4 for each partition of $k=1, 2, 3$ and 4.

Example 3.4

The coefficients $\tilde{b}_{\kappa, \tau}$ can easily be obtained by using the contents of table 2.2. The obtaining of $\tilde{b}_{\kappa, \tau}$ in the expansion of $\tilde{C}_{(21)}(D_A^1)$ for $p=4$ will be shown here.

From table 2.2 follows for $p=4$ that

$$\begin{aligned}
 (3.10) \quad \tilde{c}_{(21)}(D_A^1) &= 2(\tilde{a}_1^2 \tilde{a}_2 + \tilde{a}_1^2 \tilde{a}_3 + \tilde{a}_1^2 + \tilde{a}_2^2 \tilde{a}_1 + \tilde{a}_2^2 \tilde{a}_3 + \tilde{a}_2^2 \\
 &\quad + \tilde{a}_3^2 \tilde{a}_1 + \tilde{a}_3^2 \tilde{a}_2 + \tilde{a}_3^2 + \tilde{a}_1 + \tilde{a}_2 + \tilde{a}_3 + 2\tilde{a}_1 \tilde{a}_2 \tilde{a}_3 \\
 &\quad + 2\tilde{a}_1 \tilde{a}_2 + 2\tilde{a}_2 \tilde{a}_3 + 2\tilde{a}_1 \tilde{a}_3) \\
 &= 2(\tilde{a}_1 + \tilde{a}_2 + \tilde{a}_3) + 2(\tilde{a}_1 \tilde{a}_2 + \tilde{a}_1 \tilde{a}_3 + \tilde{a}_2 \tilde{a}_3) \\
 &\quad + 2(\tilde{a}_1^2 + \tilde{a}_2^2 + \tilde{a}_3^2 + \tilde{a}_1 \tilde{a}_2 + \tilde{a}_1 \tilde{a}_3 + \tilde{a}_2 \tilde{a}_3) \\
 &\quad + 2(\tilde{a}_1^2 \tilde{a}_2 + \tilde{a}_1^2 \tilde{a}_3 + \tilde{a}_2^2 \tilde{a}_1 + \tilde{a}_2^2 \tilde{a}_3 + \tilde{a}_3^2 \tilde{a}_1 \\
 &\quad + \tilde{a}_3^2 \tilde{a}_2 + 2\tilde{a}_1 \tilde{a}_2 \tilde{a}_3) \\
 &= \sum_{t=0}^3 \sum_{\tau} \tilde{b}_{\kappa, \tau} \tilde{c}_{\tau}(D_A^*)
 \end{aligned}$$

$$\begin{aligned}
 \text{where: } \tilde{b}_{(21), (0)} &= 0, \quad \tilde{b}_{(21), (1)} = 2, \quad \tilde{b}_{(21), (1^2)} = 2, \\
 \tilde{b}_{(21), (2)} &= 2, \quad \tilde{b}_{(21), (1^3)} = 0, \quad \tilde{b}_{(21), (21)} = 1.
 \end{aligned}$$

Remark 3.3

- (i) If $\tilde{c}_{\kappa}(D_A^1)$ is written in terms of the characteristic roots of $A: p \times p$ for different values of p by using the contents of table 2.2 it is clear that the coefficients $\tilde{b}_{\kappa, \tau}$ remain the same, independently of the values of p , for given partitions κ of k and τ of t , $t \leq k$.
- (ii) It is important to note that the coefficients $\tilde{b}_{\kappa, \tau}$ are not identical to the coefficients $b_{\kappa, \tau}$ tabulated in Khatri and Pillai ((1968), p. 224-225) for $A: p \times p$ a real symmetric matrix.

TABLE 3.4

The value of $\tilde{b}_{k,\tau}$

k	κ	τ											
		0	1	1^2	2	1^3	21	3	1^4	21^2	2^2	31	4
1	1	1	1										
2	1^2		1	1									
	2	1	1		1								
3	1^3			1		1							
	21		2	2	2		1						
	3	1	1		1			1					
4	1^4					1		1	1				
	21^2			3		1	$\frac{3}{2}$			1			
	2^2				2		1				1		
	31		3	3	3		$\frac{3}{2}$	3				1	
	4	1	1		1			1					1

REFERENCES

- CONSTANTINE, A.G. (1966). The distribution of Hotelling's generalized T_0^2 . *Ann. Math. Statist.*, 37, 215-225.
- FARREL, R.H. (1980). Calculation of complex zonal polynomials. *Multivariate analysis - V*. Edited by Krishnaiah, P.R. Amsterdam: North-Holland.
- GUPTA, A.K. (1970). Evaluation of the coefficients $g_{X,\eta,i}^s$. *Mathematica*, 12, 299-304.
- JAMES, A.T. (1964). Distributions of matrix variates and latent roots derived from normal samples. *Ann. Math. Statist.*, 35, 475-501.
- JOHNSON, N.L. and KOTZ, S. (1972). *Distributions in Statistics: Continuous multivariate distributions*. New York Wiley.
- KHATRI, C.G. and PILLAI, K.C.S. (1968). On the non-central distributions of two test criteria in multivariate analysis of variance. *Ann. Math. Statist.*, 39, 215-226.
- MACDUFFEE, C.C. (1946). *The theory of matrices*. New York: Chelsea.